

Nguyen Dinh Duc

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SUMMARY

Robotics and Artificial Intelligence student with hands-on experience in Retrieval-Augmented Generation (RAG), multi-agent AI systems, and recommendation engines. Skilled in Python, PyTorch, FastAPI, LangGraph, and modern LLM frameworks. Developed end-to-end AI applications from model development to deployment and seeking an AI Internship to contribute to real-world machine learning and generative AI projects.

EDUCATION

HaNoi University of Industry

Bachelor of Robotics and Artificial Intelligence

HaNoi, VN

2023 – Present

PROJECTS

TrafficLawRAG — Vietnamese Traffic Law Q&A System

Jan 2026 – Mar 2026

- **Tech Stack:** ChromaDB, BM25, Cross-Encoder Reranker, Groq API (Llama 3.1/3.3), Streamlit, Scikit-learn
- Built an end-to-end RAG pipeline for Vietnamese traffic law over a 6,000 QA-pair dataset, implementing 3 adaptive chunking strategies (QA-pair, sliding-window, article-boundary split) based on answer length and legal structure
- Designed a hybrid retrieval system combining BM25 (custom inverted index, pre-computed term frequencies) and dense vector search with Reciprocal Rank Fusion, achieving **100% Hit@5** and **0.983 MRR@5** on a 90-sample stratified evaluation set across 15 legal domains
- Implemented conversation memory with coreference resolution, demonstrated to recover correct retrieval context on ambiguous follow-up queries (case study: reranker relevance score improved from 1.36 to 7.26)
- Integrated a cross-encoder reranker (MiniLM-L-6) and citation extraction via regex parsing to ground every LLM answer in specific legal articles, reducing hallucination risk

Travel AI Assistant – LangGraph Multi-Agent Planning System

Apr 2026 – May 2026

- **Tech stack:** FastAPI, React, LangGraph, Qwen 3.5 Plus, Python, Docker
- Built a multi-agent travel planning system using LangGraph, coordinating specialized agents for research, itinerary generation, self-reflection, and human approval workflows
- Designed a Human-in-the-Loop architecture allowing users to review, approve, or revise AI-generated travel plans before finalization
- Developed a **6-agent orchestration system** using LangGraph Supervisor Pattern, enabling task decomposition, reflection-based self-review, and human approval before final responses
- Engineered an end-to-end travel planning platform generating detailed itineraries with budgeting, weather-aware recommendations, and feedback-driven refinement; reduced manual planning effort by **80%+** in internal testing.

Recommender-system — Multimodal Video Recommendation Engine

Jun 2026 – Jul 2026

- **Tech Stack:** PyTorch, Streamlit, NumPy, Pandas, Scikit-learn
- Built a multimodal recommender system combining collaborative filtering with visual embeddings, implementing two architectures — Multimodal NeuMF and Sequential NeuMF with a Transformer-based attention encoder — to model both static preferences and temporal user behavior
- Architected a full ML pipeline (preprocessing with negative sampling, BCE-loss training, checkpointing, evaluation) plus a Streamlit Workbench for data exploration and live Top-K recommendation demos
- Achieved **HR@10 = 0.52** and **NDCG@10 = 0.32** with the sequential model, a 35.6% relative HR@10 improvement over the non-sequential baseline (0.384), validating the value of modeling interaction order

TECHNICAL SKILLS

Languages: Python, C++, SQL

ML/DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, YOLO

LLM & Generative AI: LangChain, LangGraph, Hugging Face, Groq API, Gemini API, RAG

Databases: PostgreSQL, MongoDB, ChromaDB, FAISS

Backend & Deployment: FastAPI, Streamlit, Docker

Tools: Git, GitHub